

Pawel Sarkowicz

+1 (819) 485-0487

pawelks@duck.com

pawelsarkowicz.xyz

Languages: English (native), moderate Polish (native)

Skills: Unix/Linux (10+ years), LaTeX, public speaking

Citizenship: Canadian; Polish pending

Education

PhD, Mathematics and Statistics, University of Ottawa (Canada)

2019-2023

- Thesis: *Topics related to tensorially absorbing inclusions and algebraic K-theory of C^* -algebras*
- Topics of interest: functional analysis, operator algebras, classifying abstract quantum objects by means of simpler computable invariants
- 3.98 GPA

Master of Mathematics, Pure Mathematics, University of Waterloo (Canada)

2018-2019

- Thesis: *Exact C^* -algebras and the Kirchberg-Phillips nuclear embedding theorem*
- 4.0 GPA

Bachelor of Mathematics, Honours Pure Mathematics, University of Waterloo (Canada)

2013-2018

- Major GPA: 3.9

Experience

Mathematical Research

Undergraduate research assistant (USRA), Department of Pure Mathematics, University of Waterloo (Canada)

Winter 2018

Doctoral researcher, Department of Mathematics and Statistics, University of Ottawa (Canada)

2019-2023

Postdoctoral scholar, Department of Pure Mathematics, University of Waterloo (Canada)

2023-2025

- Conducted both independent and collaborative research in the field of operator algebras, the culmination of which has appeared in 4 refereed journal articles.
- Occasionally executed complex matrix computations, enabling more accurate understanding of abstract concepts and driving insights into future research.
- Operated 3 learning/reading seminars, and gave 15+ talks locally in others.
- Orchestrated a conference section for the Canadian Mathematical Society, and participated in 20+ other conferences.
- Refereed for 4 journals: Journal of Functional Analysis, Proceedings of the American Mathematical Society, Transactions of the American Mathematical Society, and Communications in Algebra.
- Presented 15+ research talks at international institutions, including, but not limited to, University of Oxford (UK), the Fields Institute (Canada), and the University of Houston (US).
- Mentored junior researchers at the postdoc, graduate and undergraduate level. This includes proofreading manuscripts, providing templates, having research meetings, and helping stabilize first time researchers & teachers.

University Education

Department of Pure Mathematics, University of Waterloo (Canada)

2018-2019, 2024-2025

Department of Mathematics and Statistics, University of Ottawa (Canada)

2019-2023

- Master's essay reader for 2 graduate student at the University of Waterloo.
 - Evaluated the expertise of the students, while providing valuable feedback and comments about their mathematical writing and communication skills.
- Instructor for 2 courses at the University of Ottawa: intro to linear algebra (online) and intro to proofs (in person).
 - Lectured, interacted with individual students, and built methods of evaluation.
 - Coordinated with other professors to achieve fairness and consistency across sections.
- Teaching assistant for 23 courses, 17 at the University of Ottawa and 6 at the University of Waterloo.
 - TA'd 18 distinct courses and impacted the education of 2000+ students.
 - Led problem sessions/tutorials, graded, and provided one on one sessions in tutoring centres.

Publications

- *Unitary groups, K-theory, and traces*. Glasgow Mathematical J., 2023. DOI: [10.1017/S0017089523000447](https://doi.org/10.1017/S0017089523000447)
- *Tensorially absorbing inclusions of C^* -algebras*. Canadian J. of Mathematics, 2024. DOI: [10.4153/S0008414X24000324](https://doi.org/10.4153/S0008414X24000324)
- *Polar decomposition in algebraic K-theory* (with A. Tikuisis). J. of Operator Theory, 2025. DOI: [doi:10.7900/jot.2023may29.2436](https://doi.org/10.7900/jot.2023may29.2436)
- *Universal covering groups of unitary groups of von Neumann algebras*. Studia Mathematica, 2025. DOI: [10.4064/sm240829-14-1](https://doi.org/10.4064/sm240829-14-1)
- *Kaplansky's problem and unitary orbits in matrix amplifications* (with L. Marcoux, Y. Zhang). Submitted. Preprint: [arXiv:2508.13834](https://arxiv.org/abs/2508.13834)

Awards

- University of Ottawa Admission Scholarship, Doctorate. Value: \$9000 (CAD)/year 2019-2023
- NSERC Undergraduate Student Research Award (USRA). Value: \$4,500 (CAD) Winter 2018
- University of Waterloo Merit Scholarship. Value: \$1,000 (CAD) Fall 2013

Other tidbits

- In addition to my website, my VPS hosts a mailserver, vaultwarden (bitwarden), gitea, and baikal (caldav/carddav). These are compartmentalized via docker. I plan on adding things like a self-hosted addy.io instance. I enjoy playing at being a sysadmin.
- I care deeply about privacy and security.
- I'm an avid rock climber and enjoy the outdoors.